

AQURON 2000

XỬ LÝ BÊ TÔNG TỐI ƯU



Gốc nước | Không nguy hiểm | Không chứa chất bay hơi | Không độc hại | Không chứa Isocyanates

MÔ TẢ SẢN PHẨM

AQURON 2000 là hóa chất dạng phun gốc colloidal silica hydrogel xử lý bề mặt bê tông giúp tăng cường độ bền, kiểm soát yếu tố ẩm, liên kết phân tử, và chống thấm.

AQURON 2000 xử lý từ bên trong bê tông. AQURON 2000 thích hợp cho cả hạng mục trong nhà và ngoài trời.

Các thí nghiệm độc lập đã chứng minh khi phun AQURON giúp thời gian lưu độ ẩm trong bê tông tương đương với bê tông được bảo dưỡng, và lớn hơn 90% sau 72h. Hiệu quả thủy phân từ bên trong bằng thẩm trí vượt trội các hợp chất đóng rắn màng, và vượt tiêu chuẩn đóng rắn AS3799.

ỨNG DỤNG CƠ BẢN

AQURON 2000 là dung dịch bảo vệ/trong suốt, tác dụng vĩnh viễn cho cả bê tông mới và cũ. AQURON 2000 sẽ lấp đầy bề mặt, ngăn ẩm dịch chuyển và ảo vệ đáng kể kết cấu thép. AQURON 2000 chứa chất xúc tác có độ hoạt hóa cao vốn phản ứng với kiềm tự do và/hoặc kiềm thủy phân tạo ra dạng gel silica từ bên trong điền đầy các lỗ rỗng xung quanh vật liệu cứng như sỏi, đá... Hơn nữa, kiềm được chuyển hóa hợp chất trung tính, giảm các phản ứng hóa học không mong đợi trong lòng bê tông. Tỷ trọng tăng, liên kết bền chặt hơn và bê tông sẽ vĩnh viễn chịu áp suất thủy tĩnh từ bên trong.

AQURON 2000 là sản phẩm tối ưu cho bê tông hoặc các bề mặt nền sẽ được sơn phủ vì giúp tăng độ bám dính giữa bề mặt và lớp sơn phủ. AQURON 2000 ngăn bong tróc, rạn nứt và mất kết dính do ẩm từ mao mạch hoặc các phản ứng hóa học bên trong vốn gây ra hiện tượng xà phòng hóa.

AQURON 2000 không thay đổi hình dạng bề mặt hoặc các tính chất vật lý, nên cực phù hợp cho việc lắp đặt kết cấu bê tông cũ, mới như cầu đường, cầu cảng, đường phố, đường băng sân bay, hầm, sân bãi đỗ, vỉa hè, đường giao thông, tấm bê tông ...

LỢI ÍCH LÂU DÀI

Dùng AQURON 2000 bảo vệ đáng kể bê tông không bị xuống cấp (rơi rụng ...) do đóng băng, đặc biệt đường được rắc muối hay canxi clorua trong mùa đông. AQURON 2000 loại hoàn toàn việc ẩm di chuyển từ bên ngoài đảm bảo các lớp sơn phủ bề mặt có tuổi thọ lâu hơn nhiều.

AQURON 2000 trung hòa hiện tượng ăn mòn bên trong vốn đã xảy ra do axit, ion clo, ngăn chặn các phản ứng hóa học không mong đợi. Và, AQURON 2000 làm chậm rõ đáng kể tốc độ gỉ - ăn mòn thép chịu lực. Chỉ với những dự án xây dựng ở nơi nồng độ clorua cao, mới cần sử dụng AQURON 7000.

AQURON 2000 chặn đứng hoặc ít nhất làm chậm rõ rệt bê tông xuống cấp do phản ứng kiềm-silicat. Thêm vào đó, nếu bê tông chưa bị ảnh hưởng, nếu dùng AQURON 2000 sẽ bảo vệ tuyệt đối.

THÔNG SỐ SẢN PHẨM

Dạng	Lỏng
Màu	Trong suốt
Đóng gói	can 20L / IBC 1000L
Lưu kho & Hạn dùng	Không có hạn sử dụng
Mùi	Nhẹ
Flammability	Không
VOC	Không
Điểm chớp	Không
Tỷ trọng	1.09 – 1.10
pH	12±
Độc tố	Không
Điểm sôi	110°C
Điểm đông	0°C
Tác động Môi trường	Trung tính
Hàm lượng silica tinh thể	Không
Hại do đóng băng	Không (cho phép chờ tan hoàn toàn)

HƯỚNG DẪN THI CÔNG

Chuẩn bị bề mặt

Mặt bê tông cần được làm sạch không còn bụi bẩn, lớp sơn cũ ... để AQURON 2000 có thể ngấm tối đa. Bề mặt cần được thấm ướt, nhưng không được để nước đọng trên mặt. Không thi công trên sàn đóng băng, hoặc nếu sàn sẽ đóng băng trong 12h tiếp theo. Đảm bảo bề mặt có khả năng thấm thấu.

Lượng dùng / Độ che phủ

Đề định lượng hóa chất cần dùng, nếu dùng AQURON 2000 ở tỷ lệ 4-5m²/L trên mặt bê tông có độ rỗng/độ ngấm thông thường, tuy nhiên tỷ lệ thay đổi tùy theo loại bê tông.

THI CÔNG

Thi công yêu cầu dùng máy phun có motor đảm bảo hóa chất phun ra ở dạng nguyên tử. Áp suất phun không phải là vấn đề, nguyên tử hóa là yêu cầu bắt buộc.

AQURON 2000 được phun đến khi bề mặt bão hòa (thường là khi phun đủ liều lượng), đó là thời điểm hóa chất bắt đầu chảy tràn, phun vài lớp mới đủ tỷ lệ yêu cầu.

Bắt đầu các bước thi công tiếp theo khi AQURON 2000 đã ngấm hoàn toàn (lớp mặt lớp bóng). Không để lớp phun đầu tiên bị khô; phun tiếp lớp sau khi lớp phun trước vẫn còn ướt. Nếu cần, có thể làm ướt bề mặt bê tông trước khi phun hóa chất.

Phun từ điểm thấp nhất. Bề mặt đứng cần phun từ dưới lên trên. Phun chồng các lớp xen kẽ 25 - 30% để đảm bảo phun toàn bộ bề mặt.

SƠN PHỦ & HOÀN THIỆN BỀ MẶT

Nếu bề mặt còn thi công lớp sơn phủ, cần chờ ít nhất 72hr sau khi phun, chuẩn bị bề mặt theo quy trình chuẩn.

Thi công AQURON 2000 không yêu cầu phải mài bề mặt bê tông. Nếu cần bề mặt bê tông có độ thẩm thấu, chỉ cần thực hiện bước chuẩn bị bề mặt như trên.

Tất cả các loại sơn phủ, chất kết dính được sản xuất để thi công trên mặt bê tông đều tương thích với bề mặt đã được xử lý AQURON.

NHIỆT ĐỘ

Không thi công trên sàn đóng băng, hoặc nếu sàn sẽ đóng băng trong 12h tiếp theo.

CẢNH BÁO

Rửa sạch kỹ ngay lập tức nếu hóa chất dính lên mặt kính, đồng hồ. Không để hóa chất tự khô, rửa ngay bằng nước. Nhôm bóng có thể bị mờ nếu tiếp xúc với hóa chất, kim loại không ảnh hưởng. Không phun lên mặt đóng băng, hoặc nếu khả năng đóng băng trong 12h tiếp theo.

SỨC KHỎE & AN TOÀN

Tránh tiếp xúc hay kích ứng da, nên đeo gang và kính bảo hộ. Trong khu vực kín, nên đeo khẩu trang để tránh ảnh hưởng đến hô hấp. Nếu thi công trần, đeo kính bảo vệ mắt. Tránh tiếp xúc với kính mắt, đồng hồ, vì AQURON 2000 có thể làm mờ kính và xin nhôm nếu để khô. Rửa sạch công cụ bằng nước xà phòng. Không bao giờ để lại hóa chất không dùng hết trong dụng cụ, đọc hướng dẫn nhà sản xuất.

BẢO HÀNH

Chất lượng sản phẩm: Markham Distributing Limited ("Seller") đảm bảo rằng nếu sản phẩm cung cấp bị lỗi, Bên bán sẽ thay thế hàng mới hoặc hoàn lại tiền. Markham không chịu trách nhiệm nếu khách hàng không sử dụng đúng cách AQURON 2000.

Bảo hành chống thấm (bê tông mới): 15 năm nếu thi công đúng quy trình, do công nhân được đào tạo tiến hành.

Bảo hành cho sàn bê tông mới: nếu tuân theo MARKHAM 5 uy tắc & 8 cảnh báo, có thể được bảo hành 15 năm.

10/2021.1

AQURON 2000

PHƯƠNG PHÁP XỬ LÝ

TỐI ƯU CHO BÊ TÔNG.



Gel ưa nước ngấm sâu, kiểm soát yếu tố ẩm, chống thấm từ bên trong, bao phủ và tăng tuổi thọ công trình





AQURON 2000 – HÓA CHẤT CHE PHỦ NGẤM SÂU BÊ TÔNG, CHỐNG THẤM VÀ KIỂM SOÁT YẾU TỐ ẨM TRONG BÊ TÔNG GIÚP KÉO DÀI TUỔI THỌ CÔNG TRÌNH

PHƯƠNG PHÁP XỬ LÝ TỐI ƯU CHO BÊ TÔNG

AQURON 2000 là sản phẩm dạng phun, gốc colloidal silica hydrogel xử lý cho cả bê tông cũ và mới. Lý tưởng để kiểm soát yếu tố ẩm trong lòng bê tông nền ở hàng mục ngầm như sàn hầm, vách hầm cũng như hạng mục nổi như mái Có thể sử dụng để cải tạo, kéo dài tuổi thọ công trình. Thích hợp cho hạng mục trong và ngoài nhà.

KIỂM SOÁT YẾU TỐ ẨM

Do áp lực tiến độ xây dựng, hoạt động thi công mặt sàn luôn phải diễn ra khi khối bê tông chưa được thủy hóa hoàn toàn. Kết quả là tăng đáng kể độ rủi ro do ẩm dịch chuyển, gây ra thất bại khi sử dụng các loại chất kết dính, các loại sàn trải vinyl bubbling, và các vật liệu dán sàn như epoxy, polyurethane

AQURON 2000 “làm bất động” yếu tố ẩm trong bê tông, hoàn toàn loại bỏ rủi ro do ẩm dịch chuyển và cho phép thi công mặt sàn mới sau 28 ngày hoặc chỉ sau 72hr với bê tông cũ.

BẢO VỆ/CHỐNG THẤM

AQURON 2000 ngấm sâu (tới 150mm) vào nền bê tông bằng liên kết hóa học, cố định ẩm bên trong và ngăn ẩm từ bên ngoài.

Nguyên nhân chính gây xuống cấp bê tông đều liên quan đến sự di chuyển của yếu tố ẩm, và các tạp chất liên kết với ẩm, bên trong lỗ rỗng của bê tông. AQURON 2000 chống thấm vĩnh viễn, ngăn các hiện tượng trên và không cần phải thi công bảo dưỡng trong tương lai.

CÁC QUAN ĐIỂM KHÁC

AQURON 2000 phản ứng trong lòng bê tông; nên không tạo lớp màng nhìn thấy trên bề mặt.

Lấp kín lỗ rỗng bê tông tới 150mm trong lòng bê tông, và không bị bong tróc do xe cộ đi lại, quá trình mài phẳng. Hàn kín vết nứt đã ổn định có độ rộng tới 0.5mm.

Tăng cường bê tông, tránh rủi ro xuống cấp do phương tiện giao thông, và thích hợp với bề mặt ngoài trời.

Hoàn toàn tương thích với các loại sơn phủ, chất kết dính cho mặt bê tông; không tạo ra yếu tố nào gây hại cho bê tông.

CƠ CHẾ PHẢN ỨNG

Công nghệ cốt lõi là colloidal silica hydrogel – gel ưa nước gốc silica dạng keo tụ. Khi phun lên mặt bê tông, phân tử silica kích thước nano phản ứng với calcium hydroxide trong lỗ rỗng chứa nước để thẩm thấu vào lỗ rỗng trong lòng bê tông, ngấm sâu tới 150mm tạo ra dạng gel ưa - hydrogel nước gốc silica, bền vững, polymer hóa, điền đầy lỗ rỗng, xóp trong lòng bê tông.

Dạng hydrogel này theo thời gian sẽ đạt độ cứng tương tự Calcium Silicate Hydrate (cement paste). Chức năng quan trọng nhất, ở bất kỳ tuổi bê tông nào, giữ nhiệm vụ ngăn ẩm dịch chuyển trong các lỗ rỗng của bê tông. Một khi phản ứng diễn ra, hydrogel được tạo ra, trạng thái là vĩnh cửu không đảo ngược hoặc mất đi, nó trở thành một phần của bê tông.

Colloidal silica hydrogels vốn ở kích thước nano và có thể phản ứng trong mao mạch cỡ nano của bê tông. Lỗ rỗng phổ biến của bê tông là khoảng 50 nm, còn kích thước colloidal silica trong khoảng 5-20 nm. Để dễ so sánh, sợi tóc có kích thước 100,000 nm.

CÁCH THI CÔNG



AQURON 2000 thi công bằng cách phun, theo định mức 4.5m² /L, tùy theo độ ngấm của bê tông. Máy phun cần áp suất đủ để tạo ra hạt phun sương.

AQURON 2000 dùng được cho bê tông mới và cũ. MARKHAM khuyến cáo nên dùng AQURON 2000 với bê tông mới, ngay trong quá trình đóng rắn, để hỗ trợ quá trình đóng rắn và tối ưu hóa các lợi ích.

CÔNG DỤNG

ĐÓNG RẮN

Hỗ trợ đóng rắn : Tương đương 14 ngày ngâm/phun nước bảo dưỡng
Đạt tiêu chuẩn giữ ẩm cho bê tông AS3799
Tiết kiệm thời gian và chi phí đóng rắn, giúp kiểm soát yếu tố ẩm trong lòng bê tông

CHỐNG THẤM

BS EN12390 - Part 8: Thí nghiệm độ ngấm nước dưới áp suất nước sâu 50m cho thấy cải thiện 90% so với mẫu đối chứng
DIN 1048 - áp suất nước nén ở độ sâu 10m – cải thiện 72% so với mẫu đối chứng
GWT Field Permeability – tối thiểu 75% so với mẫu đối chứng

KHÓA ẨM – NGĂN ẨM DỊCH CHUYỂN

Khóa ẩm bằng cách tạo ra hydrogel. Cho phép thi công sàn ngay sau đó, theo tiêu chuẩn AS1884-2012, bỏ qua RH% của bê tông. Tuân theo các quy chuẩn AS1884-2012.
Conforms to moisture suppressant requirements of AS1884-2012.
Để đảm bảo thi công đúng, nên tuân thủ quy tắc thi công của "MARKHAM Flooring Installation Guide" (5 Quy tắc & 8 Cảnh báo).

TĂNG CỨNG & CHỐNG BỤI

Thí nghiệm chịu mài mòn Chaplin cho thấy cải thiện rõ rệt (tăng 40% khả năng chịu mài mòn) ở bê tông đã phun AQURON 2000
Thí nghiệm đo độ bụi trên mặt bê tông cho thấy kết quả (giảm 30% bụi) khi dùng AQURON 2000

LẤP ĐẦY & VÁ VẾT NỨT

Hàn gắn vết nứt độ rộng tới 0.75mm width – cho thấy hiệu quả
Tự vá – vết nứt ổn định có độ rộng tới 0.5mm

THÂN THIỆN BỀN VỮNG

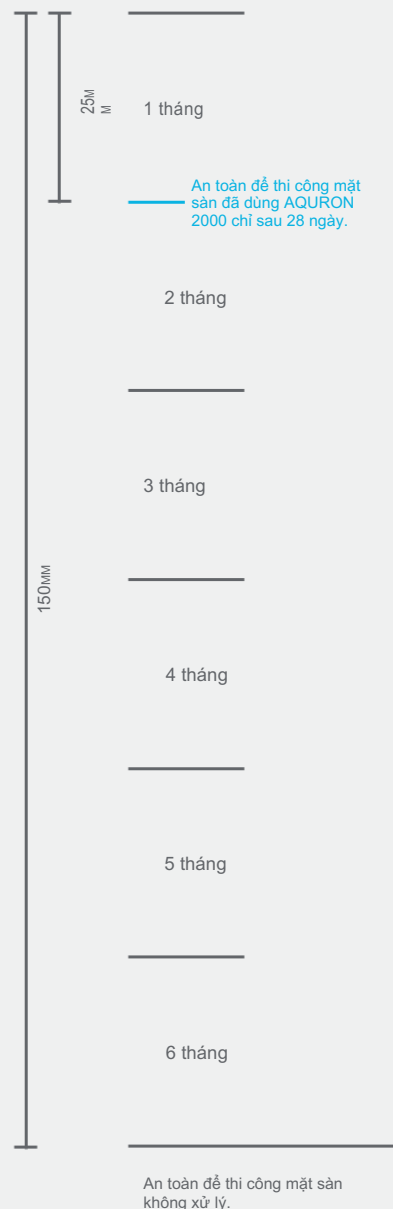
Quá trình sản xuất không tạo ra sản phẩm phụ độc hại như các sản phẩm silica khác
Giúp cấu trúc bê tông bền lâu hơn, giảm yêu cầu bê tông mới, nghĩa là giảm bớt khói thải đốt than trong quá trình sản xuất xi măng

AN TOÀN

Hàm lượng VOC = 0. An toàn cho người và động vật, sinh vật biển.
Nước sinh hoạt – được chấp nhận sử dụng - AS4020.
Bất cháy – sản phẩm gốc nước, không cháy nổ.

BẠN CÓ BIẾT – TỐC ĐỘ KHÔ CỦA BÊ TÔNG LÀ 25MM /THÁNG?

Thời gian khô của nền bê tông



SẢN PHẨM PHỤ TRỢ

Để đảm bảo hiệu quả chống thấm cao nhất, có thể phun AQURON 2000 và kết hợp với phụ gia dạng trộn và các loại thanh trướng nở.

AQURON 2000 + CONQOR B50 phụ gia chống thấm + **CONQOR 47B** thanh cản nước trướng nở.

TALK TO US. WE KNOW WHAT WORKS WHERE, AND WHAT DOESN'T.



1800 693 694 Australia New
0800 693 694 Zealand United
0333 456 5758 Kingdom
0904 128 909 Vietnam



DEE WHY PCYC CAR PARK

DEE WHY, NSW

Chống thấm bê tông ngoài trời chịu mọi thời tiết. Hoàn thành giữa năm 2016.



HARINGTON STREET TRANSPORT HUB

TAURANGA, NEW ZEALAND

Năm 2022, AQURON 2000 được dùng cho sàn nhà để xe bỏ hoang trong nhiều năm



WOOLWORTHS CORNUBIA

CORNUBIA, QUEENSLAND

Bảo vệ nền bê tông trước khi phủ sàn vinyl, xử lý yếu tố ẩm. Hoàn thành 2016.



PORTS OF AUCKLAND BLEDISLOE CAR PARK

AUCKLAND, NEW ZEALAND

AQURON 2000 được dùng cho sàn nhà để xe 33.000 m2. Hoàn thành 2018.



GLEBE ISLAND WHARF#1

GLEBE ISLAND, NSW

Bến cảng, ngoài trời, chịu muối biển, xử lý vết nứt, chống thấm, tăng độ bền, kéo dài tuổi thọ công trình. Hoàn thành 2019



NZ CULTURAL & HERITAGE WAREHOUSE

TIMARU, NEW ZEALAND

AQURON 2000 dùng để chống thấm và kéo dài tuổi thọ công trình vốn được xây từ 1874.



MARKHAM
adding life to concrete



Mô tả sản phẩm

AQURON 2000 là dung dịch gốc nước trong suốt, phân tử liên kết với bề mặt bê tông cho khả năng chống thấm, chịu mài mòn, hỗ trợ đóng rắn.

Quy trình thi công

ĐIỀU KIỆN THI CÔNG

1. Không thi công Aquron2000 nếu nhiệt độ không khí hoặc nhiệt độ nền dưới 2,8°C hoặc dự báo thời tiết nhiệt độ thấp hơn 2,8°C trong 6h tiếp theo.
2. Nếu nhiệt độ mặt bê tông quá nóng, phun đủ nước để làm giảm nhiệt độ.
3. Trong điều kiện gió to cần lưu ý khi thi công, nếu để Aquron2000 khô, nền ánh kim hoặc bóng như gương. Cần dùng nước phun rửa trước khi Aquron2000 bị thổi khô trên mặt bê tông.
4. Do Aquron2000 là sản phẩm gốc nước, cần tiếp xúc nền bê tông nên cần làm sạch các lớp sơn phủ hay dầu mỡ bám trên bề mặt.
5. Không pha loãng
6. Sau khi phun Aquron2000, chờ 28 ngày nếu là bê tông mới, chờ 72h nếu là bê tông cũ trước khi sơn phủ.
7. Aquron2000 hàn gắn bên trong bê tông, không hàn gắn vết nứt to, nứt kết cấu.

QUY TRÌNH THI CÔNG

1. Nếu là bê tông cũ, phun Aquron2000 bằng máy phun áp suất cao (0.19,25⁰)
2. Làm ẩm bề mặt trước khi phun, quét sạch vũng nước, giữ ẩm bề mặt cho đến khi phun Aquron2000.
3. Định lượng phun Aquron2000 khoảng 5m²/L trên bề mặt bê tông có độ rỗ, độ ngấm thông thường. Để đạt yêu cầu che phủ như trên, thường phun ít nhất 2 lớp. Lớp 1 phun theo 1 chiều đồng nhất, chờ 10-60ph, không để bề mặt khô rồi mới phủ lớp 2 theo chiều vuông góc với lớp 1.
4. Khi phun, giữ khoảng cách đầu phun và bề mặt bê tông khoảng 20-30cm. Có thể dùng thiết bị hỗ trợ để dàn trải đều hóa chất trên bề mặt.
5. Bề mặt phải được phun đầm Aquron2000, nhưng không để lại vũng, dùng chổi gạt vũng 15-30ph sau khi phun, không được để vũng đọng khô.
6. Luôn phun hóa chất từ vị trí thấp nhất lên vị trí cao. Nếu phun lên tường, phun từ dưới lên trên theo chiều thẳng đứng.
7. Trước khi thi công sơn sàn đã phun Aquron2000, tuân theo các quy tắc Aquron 5 QUY TẮC & 8 CẢNH BÁO. Kiểm tra Aquron đã ngấm vào bê tông, làm sạch dầu mỡ, muối Phun và làm sạch bề mặt bằng nước trước khi thi công sơn sàn, chất bám dính.
8. Nếu dùng cho tường, sàn chịu áp suất thủy tĩnh. Ít nhất phun 2 lớp, lớp thứ 2 phun ngay sau lớp 1. Không để lớp 1 khô trước khi phun lớp 2.

THỜI GIAN KHÔ: 2h ở điều kiện 21°C.

CẢNH BÁO: Nếu dính hóa chất lên kính hoặc vật liệu nhôm, cần rửa sạch ngay.

VỆ SINH DỤNG CỤ THIẾT BỊ: rửa sạch bằng nước sau khi thi công.



Australia 1800 693 694	www.markhamglobal.com.au
New Zealand 0800 693 694	www.markhamglobal.com
United Kingdom 03334 56 57 58	www.markhamglobal.co.uk

Liên hệ Nhà phân phối: **ITSV CO., LTD**

Công ty TNHH Dịch vụ và Thương mại Quốc Tế Việt

40/20 Cát Linh, Đống Đa, Hà Nội. Tel: **0904128909** - E: **chau@itsv.com.vn**





CÔNG TY TNHH CHỨNG NHẬN VÀ KIỂM ĐỊNH CHẤT LƯỢNG OPACONTROL
TRUNG TÂM THÍ NGHIỆM KIỂM ĐỊNH CHẤT LƯỢNG LAS-XD 635

Địa chỉ: số 40, ngõ 282 đường Kim Giang, phường Đại Kim, quận Hoàng Mai, HN.

Chi nhánh: số 622/1 đường Cộng Hòa, phường 13, quận Tân Bình, TP HCM

Địa chỉ PTN: số 25-27 Trương Định, phường Trương Định, quận Hai Bà Trưng, Hà Nội

Tel: 024.2206.1628 **email:** lab635@lab635.com **web:** www.opavietnam.vn

HĐKT : N/A

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

Số/No: 01.1701/23/TNVL

Trang 01/01

1. Tên đơn vị/Client: CÔNG TY TNHH TẬP ĐOÀN XÂY DỰNG DELTA/DELTA Corporation.

Địa chỉ/Address: 81 Lạc Trung, Hai Bà Trưng, Hà Nội/ *81, Lac Trung street, Hai Ba Trung district, Ha Noi city.*

2. Công trình/Project: N/A.

3. Loại mẫu/Kind of sample: Aquaron 2000.

4. Số lượng/Quantity: 1L

5. Ký hiệu mẫu/Sample: 01.1701/23/TNVL.

6. Ngày nhận mẫu/Date of receiving: 23/11/2022 – November 23, 2022.

7. Ngày cấp kết quả/Date report: 17/01/2023 – January 17, 2023.

8. Thiết bị thử nghiệm/Equipments: Máy trộn bê tông, cân kỹ thuật. máy thử mài mòn, tủ sấy, máy nén bê tông, máy thử độ thấm nước, côn thử độ sụt.../ *Concrete mixer, precision balances, abrasion testing machine, drying oven, concrete compression machine, Concrete water impermeability apparatus, slump cones set...*

9. Người gửi mẫu/Sampler: Mẫu do cơ quan yêu cầu thử nghiệm lấy và gửi đến/ *Samples are taken and sent by the client.*

KẾT QUẢ THỬ NGHIỆM
TESTING RESULTS

TT No	Tên chỉ tiêu Specification	Đơn vị Units	Kết quả Results		Phương pháp thử Test method
			Mẫu đối chứng (Controlled sample)	Mẫu Aquaron 2000 (Aquaron 2000 sample)	
1	Độ hút nước/Water absorption	%	5,8	5,4	TCVN 3113:2022
2	Cường độ nén/Compressive strength	MPa	31,6	40,8	TCVN 3118:2022
3	Độ chịu mài mòn/Abrasion	g/cm ²	0,33	0,25	TCVN 3114:2022
4	Độ thấm nước dưới áp lực thủy tĩnh 150Kpa trong 7 ngày /Water impermeability of 150kPa for 7 days	Thấm /Không thấm – Penetration/ No penetration	-	Không thấm/No penetration	BS EN 14891 TCVN 3116:2022

Thí nghiệm viên

Tested by

Phạm Đức Minh

Trưởng phòng thí nghiệm

Chief of Laboratory



Nguyễn Văn Quảng

Giám Đốc

Director



Nguyễn Thanh Tú

Ghi chú/Note:

- Tên mẫu, ký hiệu mẫu, tên cơ quan gửi mẫu và công trình sử dụng được ghi theo tài liệu đi kèm mẫu/ *Sample name, sample symbol, name of the agency sending the sample and the work used are recorded according to the documents accompanying the sample.*
- Không được sao chép từng phần (được sao chép toàn bộ) phiếu kết quả này khi chưa được sự đồng ý bằng văn bản của OPAcontrol/ *This result sheet may not be partially copied (reproduced in whole) without the written consent of OPAcontrol..*
- Kết quả thử nghiệm chỉ có giá trị trên mẫu thử nhận được/ *Test results are valid only on samples received.*



Commercial In Confidence

USE OF AQURON 2000 AS A CURING AGENT

For:

MARKHAM DISTRIBUTING LTD

Ref: J067838

Prepared: **Roger Scerri**

Dated: **11th May 2006**



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This report is Privileged and Confidential and has been prepared for the Client for the purposes of the Client only .

4 Conclusion

The test results clearly show that using Aquaron 2000 as a curing compound yields properties which are at least equivalent and at best superior to the wrapping method described as "wet curing".

Curing using Aquaron 2000 improves achieved compressive strength and more closely approaches the maximum 28 day compressive strength than cores taken from a "wet" cured or wrapped slab. Long term core strength measurements show that the use of Aquaron 2000 as a curing compound with subsequent air curing achieves similar strengths to "wet" curing.

Drying shrinkage is dramatically reduced using Aquaron 2000. This is due to the pore blocking action of the Aquaron 2000 dramatically slowing drying of the shrinkage specimens.

Curing using Aquaron 2000 improves abrasion resistance and core compressive strengths more closely approach the maximum 28 day cylinder compressive strengths than cores taken from a "wet" cured or wrapped slab. Later age abrasion testing shows that Aquaron 2000 as a curing compound with subsequent air curing achieves similar abrasion resistance to "wet" curing.

The physical appearance of the Aquaron 2000 treated slab does not exhibit the efflorescence shown on the slabs covered with polyethylene sheets.

The testing shows that curing using Aquaron 2000 and air drying is at least equivalent to "wet" curing by wetting the concrete and wrapping in polythene for 14 days with the added advantage of a better overall appearance and quicker access to the slab.

REGULATION 31(4)(A) OF THE WATER SUPPLY (WATER QUALITY) REGULATIONS 2016 SI No 614
& THE WATER SUPPLY (WATER QUALITY) REGULATIONS 2010 SI No 994 (W.99)(AMENDMENT
REGULATIONS 2016 No 410(W.128))

APPROVAL CONFIRMATION

Product name/designation	Aquron 2000
Approval Holder (Name & Address)	MG Solutions (UK) Ltd 3 Sandiacre Road, Stapleford, Nottingham, NG9 8EX
Instructions for Use	Aquron 2000 Colloidal Silicate for waterproofing of concrete. Instructions For Use (IFU) Issue date 21/08/2017; Issue Number 1.1
DWI Ref Number	DWI 56.4.1291

The Secretary of State for Environment, Food and Rural Affairs in respect of relevant water suppliers¹ whose area of supply is wholly or mainly in England, and the Welsh Ministers in respect of relevant water suppliers whose area of supply is wholly or mainly in Wales, in exercise of their powers under Regulation 31(4)(a) of the Water Supply (Water Quality) Regulations 2016 and Regulation 31(4)(a) of the Water Supply (Water Quality) Regulations 2010 (Wales) respectively hereby approve the introduction of the product detailed above, in accordance with the following Conditions of Use & Approval.

CONDITIONS OF USE & APPROVAL

The above named product is approved for application in England and Wales² by a water supply company for public water supply purposes subject to the following conditions:

1. Water undertakers and water supply licensees shall be provided with a copy of the Instructions for Use (IFU) detailed above and use of the approved product must be in accordance with the IFU.
2. The Drinking Water Inspectorate (DWI) must be notified, in advance and in writing, by the approval holder detailed above, in respect of any **further** change(s) in –
 - a) The Instructions for Use
 - b) The formulation of the approved product, including change in source or identity of raw materials
 - c) The manufacturing process, including location of manufacture
 - d) The designation of the approved product
 - e) The name, address or ownership of the organisation holding the approval

Failure to notify such further changes will result in approval being withdrawn.

3. The producer shall ensure that the product is tested for conformity with its formulation, and the source or identity of its raw materials, at such intervals and by such persons, as may be determined by the Authorities. The results of such testing shall be sent to DWI.
4. The use of either the Drinking Water Inspectorate or DEFRA logos, in respect of any approved product (including on the product or in editorial or advertising/trade copy) is not permitted.
5. The following changes to any product specific condition(s) apply – **None**

Signed by authority of the Secretary of State
and the National Assembly for Wales

Date of issue: 03/11/2017
Date of Expiry: 02/11/2022



Marcus Rink
Chief Inspector of Drinking Water
England and Wales

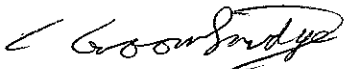
¹ Relevant water suppliers shall include water undertakers, water supply licensees and inset appointees.

² Separate approval arrangements apply in Scotland and Northern Ireland

ACKNOWLEDGEMENT OF RECEIPT OF APPROVED PRODUCT CONFIRMATION & CONDITIONS OF APPROVAL

Product name/designation	Aquron 2000
Approval Holder (Name & Address)	MG Solutions (UK) Ltd 3 Sandiacre Road, Stapleford, Nottingham, NG9 8EX
Section in the "List of Approved Products"	C.6 Sealant & Repair Materials
DWI Ref Number	DWI 56.4.1291

Please carefully read the accompanying Product Approval document, together with the accompanying letter. If there is any error please notify us as soon as possible.
Please confirm, by ticking the boxes below and then signing and returning this form, the safe receipt of this documentation and your agreement to the conditions of use and approval set out in it.

<i>On behalf of the approval holder I confirm -</i>	Tick here
The safe receipt of the approval documentation and <i>EITHER</i> – • All details are correct and no changes are required <i>OR</i> • Corrections concerning the approval product are needed and I am returning to you, a corrected copy of the "Product Approval" for amendment and reissue	☒ OR ☐
Our agreement to comply with the conditions of use and approval set out in the "Product Approval", including – • Ensuring that a copy of the appropriate Instructions for use is always provided with the approved product as sold • Ensuring that the Drinking Water Inspectorate is informed of any further proposed change(s) to the approved product • Not using either the Drinking Water Inspectorate or the DEFRA logos in association with the approved product.	☒ ☒ ☒
We understand that the approval of this product under Regulation 31 relates only to potential effects on drinking water quality. It does NOT imply fitness-for-purpose or an endorsement of the product, and no general claim of "Drinking Water Inspectorate (DWI) approval" should be made.	☐
Signed 	
Name CARL GROOMBRIDGE	Date 06/11/2017

IMPORTANT NOTES

1. Upon receipt of this completed acknowledgement we will notify the Scottish Government and the Department for Regional Development, Northern Ireland of the approval; these bodies are responsible for issuing their own approvals for use of the product in Scotland and Northern Ireland, under Regulation 33 of the Water Supply (Water Quality) (Scotland) Regulations 2014 and Regulation 30 of the Water Supply (Water Quality)(Amendment) Regulations (Northern Ireland) 2009, respectively.
3. If we do not receive this signed acknowledgement with acceptance by you of all conditions of approval within 6 months, the acceptance of the change(s) to this approved product will lapse automatically.
4. Please return the signed form to – Regulation 31 Enquiries, Area 7e, 9 Millbank, c/o Nobel House, 17 Smith Square, London SW1P 3JR, U.K.



guardians of drinking water quality

DRINKING WATER INSPECTORATE

Area 7e, 9 Millbank
c/o Nobel House
17 Smith Square
London SW1P 3JR

Enquiries: 0300 068 6400

E-mail: Joanna.hunt@defra.gsi.gov.uk
DWI Website: <http://dwi.defra.gov.uk>

Date 03/11/2017

Dear Carl and Simon

**APPROVAL GIVEN UNDER REGULATION 31(4)(a) OF THE WATER SUPPLY (WATER QUALITY)
REGULATIONS 2016 SI No. 614 & THE WATER SUPPLY (WATER QUALITY) REGULATIONS
2010 SI No.994 (W.99) (AMENDMENT REGULATIONS 2016 No 410(W.128))**

We are pleased to enclose the "Product Approval Confirmation" for your product for use in England and Wales. The Drinking Water Inspectorate advises both the Secretary of State for Environment, Food and Rural Affairs (DEFRA), and the Welsh Ministers (collectively referred to as "the Authorities") on the approval of products used in the provision of public water supplies.

This approval is given on the basis that there is no objection on health grounds to the use of this product in the provision of public water supplies. Approval does not include consideration of fitness for purpose of the product, and cannot be taken as a favourable assessment of the performance or technical merits of it. This approval relates only to the product described in the "Product Approval Confirmation" when it is used in accordance with the "Conditions of use and approval" specified in that document; failure to comply with these or any other condition of approval could lead to revocation of approval. This approval is valid for 5 years.

You are **not** authorised to use either the Inspectorate's logo or that of DEFRA in advertisements of any other literature relating to the approved product. Since this approval is given by the Authorities, you should not refer to approval by the Drinking Water Inspectorate. Advice on acceptable claims for your approved product is given on our website - <http://dwi.defra.gov.uk/drinking-water-products/faqs/FAQ2.pdf>.

Finally it is important that you complete, sign and return the enclosed "Acknowledgement of Receipt of Product Approval Confirmation and Conditions of Use" as soon as possible, since we cannot add your product to the "List of Approved Products for use in Public Water Supply in the United Kingdom" until this is received. Upon receipt full details of your product will be passed to the Scottish Government and the Department for Regional Development, Northern Ireland, so that they can review your product for approval in their respective parts of the United Kingdom.

Yours sincerely

Joanna Hunt
Inspector

**Markham Australasia
43 Niven Street
Napier**



Certificate of Testing

Issued: April 2016

This certification of cover testing is based on a series of permeability pressure tests undertaken by G-Group over a period from April to November 2015 as follows:

- Standard compression test moulds were cast of various concrete strengths
- Concrete strengths were 30MPa and 50MPa
- The concrete cylinders were cured at 23c and RH 50% to various ages before treatment
- Prior to cutting of the cylinders they were sealed on their curved surface to prevent moisture loss under pressure
- Samples were slices diamond saw cut at 30 and 60mm thickness
- Control samples were used to provide direct treated comparative findings
- The top surface of the treated samples was sprayed with Aquaron 2000 in accordance with the manufacturer's technical data sheets
- Pressure applied to the sprayed surface varied between 50 and 110psi (approx. 25 and 65m head)
- Different time periods were tested of between 10 minutes and greater than 24 hours

The penetrating hydrogel ability to firstly penetrate the concrete and then to immobilise moisture movement internally is being investigated under this test regime.

The penetration pressure rig measures the moisture going into both the treated and control samples providing a direct comparative view of the treatment. Some samples were after treatment cut to remove the top 5mm to simulate abrasion, typical concrete surface cracking and chipping.



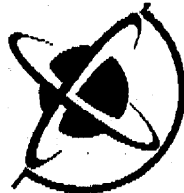
The findings are certified as below.

- ***The permeability pressure tests showed that the Aquaron2000 spray applied treatment***
 - ***modified the internal concrete matrix to reduce the concrete's ability to allow mobilised water movement within the concrete pore and interstitial spaces***
 - ***this reduced the volume of new moisture entering under pressure by a factor of between 3 and 8***
 - ***with results showing a grouping at a factor of 4 for all pressure ranges***
 - ***and removing some of the treated surface had little impact on the findings***
- ***This supports earlier testing that suggests a conservative factor of 2 for situations where inadequate cover may have eventuated following steel movement during construction. That is a cover of 40mm treated is equal to a cover of 80mm untreated***

The findings of this test regime were presented to the Coast and Marine Structures 2016 Conference in Brisbane in March 2016.



Murray Gilbertson
Managing Director
BE (Hons) Civil, BS Dip, MBA (Otago)
G-Group Consulting Limited
Mobile: +64 274 814 684
Email: mggilbertson@gmail.com



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**GEOLOGICAL
 & NUCLEAR
 SCIENCES**
 Limited

facsimile
 NS-BJB

FAXED

To Graham Smith

Fax 061972 4121543

Company Markham Distributing

Fax Number 06 878 5143

From Bernard Barry (email b.barry@gns.cri.nz)

Date 29 January 1999

Time

If you do not receive all 2 pages, please telephone immediately.

This facsimile message is a confidential message between the Institute of Geological & Nuclear Sciences Limited and the intended recipient. If it has been received by you in error, please notify us by telephone immediately and return the original facsimile to us by mail at the address shown below. Thank you for your cooperation.

Dear Graham

RESULTS OF AQURON EXPERIMENTS

In conjunction with you I recently carried out tracer experiments involving mixing the radioactive isotope ^{125}I (iodine-125) with aquaron and applying it to concrete test cylinders. The cylinders were prepared in such a way that the applied material could only contact the cylinder surface. The remainder of the cylinder was contained within shrink wrap material. Aquaron mixtures were applied in some instances by spraying and in others by pooling in liquid of depth several millimetres on the tops of the cylinders.

The day after tracer application a radiation detector was used to monitor the tracer on top of the cylinders. Although about the same amount of tracer had been applied to each, it was found that the detector readings from the sprayed samples were about five times lower than from the pooled samples. This implied penetration of the tracer into the cylinders which had been sprayed; otherwise the sprayed samples would have been expected to give readings on average at least as high as the pooled samples.

Visual examination after several days showed evidence in some cases of liquid within the shrink wrap material. Measurement with a radiation detector confirmed that it contained some of the tracer.

The cylinders were cut in half lengthwise and scanned with a detector able to detect tracer within a few mm of the inspected surface. This indicated the presence of tracer in the cylinders, in particular within the first 50 to 100 mm from the top. The field of view of the detector (a slit about 50 mm wide) included aggregate which would not be expected to contain tracer or aquaron so it is only possible to use these results to confirm the presence of the tracer. In addition the field of view could include some cement which contained tracer and some which did not.

Five samples were taken by collecting powder from drilling, four from the cylinders to which tracer had been applied and one from a separate piece of concrete. These were counted in a sensitive detector. Two of the samples, taken from 20 and 40 mm below the surfaces of their respective cylinders, showed definite evidence of the tracer. These corresponded to regions

PO Box 31312, Lower Hutt, New Zealand, Telephone: +64-4-570 4637, Facsimile: +64-4-570 4657
 Delivery Address: 30 Gracefield Road, Lower Hutt, New Zealand

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where the scanning had shown the presence of tracer but the amounts did not correspond in the two measurements; this is probably related to the point noted above concerning variations in tracer content within the detector field of view. There was no tracer noted in the two samples from points at which the scans showed no tracer. As in scanning it cannot be guaranteed that these samples did not contain aggregate so it is not possible to use these results to assess the amount of tracer in the cylinders.

To summarise, the results showed unequivocally that the tracer was able to migrate into the interior of the cylinders. In one case the movement was over a distance of about 200 mm and liquid containing the tracer was observed between the cylinder surface and the shrink wrapping in which it was enclosed. It is my opinion that there was no path for the tracer to reach this position other than through the bulk of the cylinder, at least over the first 30 or so mm. Other measurements revealed the presence of tracer in the interior of the cylinders. The results showed no consistency between cylinders suggesting that the movement is dependent on individual cylinder characteristics. I am also not able to determine how much of the tracer penetrated the concrete.

Regards



B Barry

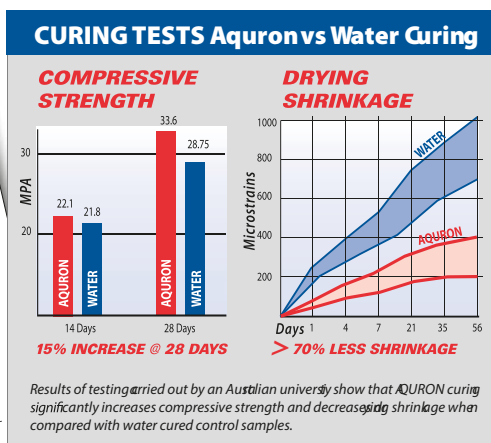
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Delivery Address: 30 Gracefield Road, Lower Hutt, New Zealand

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TOTAL P.02

AQURON has been proven to be equivalent to curing concrete for 14 days under water, and exceeds industry standards.

AQURON has a successful 12 year track record of use extensively in both New Zealand and Australian conditions, and has Australasian independent testing which confirms AQURON curing performance achieves "at least equal" results in direct comparison with WATERCURING*



*Testing has proved that AQURON "internal" curing has matched and even exceeded WATERCURING results in quality concrete, as the AQURON treatment draws in and incorporates moisture as additional hydrate formations. ASTM C309 and ANZS 3799 Standards state that WATERCURING is superior to MEMBRANE CURES which can cause "low water/cement ratio" concrete to self desiccate (i.e. starve for moisture). Refer ANZS 3799 clause D7.

- ✓ Curing Assistance
- ✓ Shrinkage Reduction
- ✓ Strength Enhancement
- ✓ Durability Enhancement
- ✓ Sheen Enhancement

WHERE FLOORCOVERINGS or COATINGS ARE TO BE INSTALLED ...



Avoid the extra cost of grinding off traditional surface curing compounds.

Traditional surface curing compounds are designed to retain moisture in the concrete.

As a result, moisture levels in the slab frequently exceed what is acceptable for safe installation of floor coverings.

As a consequence further expenditure is necessary to remove these compounds and provide moisture control later in the construction program.

Achieve permanent, cost efficient moisture control beneath flooring at time of curing, using the all-in-one AQURON 2000 CURE & SEAL, which offers guaranteed INTEGRAL MOISTURE CONTROL SYSTEM.

AQURON is a reliable, effective and convenient alternative to WATERCURING, and superior to moisture retention membranes.

Various independent testing laboratories have shown that AQURON CURING can achieve moisture retention in excess of the requirements of ASTM C309

Specify and use AQURON with confidence as a CURING METHOD, knowing that the "internal" curing action permanently incorporates moisture and directly increases achieved strengths, and reduces shrinkage.

Because of the "supply & apply" policy that Markham's offer, you have greater quality control, and you are assured that what you specify is what you get!

For more information on AQURON INTERNAL CURING
Talk to one of our team.



Innovations for Concrete & Construction

www.aquron.com.au	1800 693 694
www.aquron.co.nz	0800 693 694
www.aquron.co.uk	03334 56 57 58

GORDON SPRATT & ASSOCIATES LTD.

CONSULTING PROFESSIONAL ENGINEERS

2348 Yukon Street, Vancouver, British Columbia

V5Y 3T6

Phone (604) 872-1211

Fax (604) 872-1274

MARKHAM DISTRIBUTING

309 East Avenue Road

P.O. Box 2182

Hastings, New Zealand

(Fax: 011-646-878-5143)

Our File No. M-499-97

November 28, 1997

Attention: Mr. Graham Smith

Dear Sir:

Re: AQURON PRODUCTS

- Aquoron 2000 Concrete Preservation Treatment

- Evaluation

A review of all of our testing to date also has been done, and we find that the Aquoron 2000 Concrete Preservation Treatment product will perform as intended according to the series of ASTM tests that we have conducted. Our firm has been monitoring the effectiveness of the Aquoron 2000 Concrete Preservation Treatment used on several projects for the last 7 to 8 years. It has been found through annual testing and review of these projects, as required by the City of Vancouver Engineering Department, that the Aquoron 2000 Concrete Preservation Treatment enhanced the performance of the concrete in these projects. Our responsibility on these projects were to obtain approval from the City of Vancouver, for an alternate concrete mix design in conjunction with Aquoron 2000 Concrete Preservation Treatment for suspended parkade slabs, thus eliminating the annual maintenance required for membrane systems. We have found that with the use of the Aquoron 2000 Concrete Preservation Treatment product that the concrete tested annually in the last 7 years is virtually chloride free, of extreme low permeability and no detriment to the reinforcing steel has been recorded. All tests performed annually are to acceptable ASTM standards.

Yours truly,

GORDON SPRATT & ASSOCIATES LTD.

Per:



Don E. Lee

Technician

DEL/as

Encl.

APPRAISAL CERTIFICATE

FOR

AQURON 2000 CONCRETE PRESERVATION TREATMENT

1. PRODUCT NAME

- 1.1. Aquaron 2000 Concrete Preservation Treatment is manufactured by Aquaron Corporation, P.O. Box 758 Rock Wall, Texas, 75087 (Telephone: 972-412-4649/ Fax: 972-412-1543).

2. PRODUCT DESCRIPTION

- 2.1. Aquaron CPT-2000 is a clear sub-surface molecular bonding and waterproofing solution.
- 2.2. Aquaron CPT-2000 is a proprietary silicate base which produces a silica hydrogel below the surface and inside its matrix.

3. PRODUCT INFORMATION

3.1. Recommended For:

- a) Freshly placed concrete.
- b) Aged concrete.
- c) Curing conventional concrete.
- d) Protective sealer for pavements, docks, ramps, paving stones, precast concrete decks, stamped concrete, exposed aggregate concrete, masonry and terrazzo.
- e)** Can control water and water vapour through walls and floors as a result of hydrostatic pressure (note: all leaking joints and cracks to be properly repaired prior to the application of the product).

3.2. Features:

- a) Seals concrete from within leaving no residues or membrane on concrete surface—"non-yellowing".

- b) Is not subject to damage from heat, ultraviolet light and chemical attack.
- c) Increases concrete durability.
- d) Protects concrete from chloride ion penetration which causes corrosion of reinforcing steel.
- e) Provides moisture retention capabilities for curing fresh concrete.
- f) Both curing and sealing of fresh concrete achieved in one spraying application.
- g) Simple to apply—can be applied to vertical and horizontal surfaces of concrete with varying textures in or outdoors.
- h) Quickly “migrates into” and penetrates concretes—only needs a brief period before “back in use” or serviceability—“speeds up” new concrete surfaces for use.
- i) Provides excellent bonding for adhesives, impermeable coatings, paint, plasters, topping concrete.
- j) Prevents internal migrations and release of moisture and salts in concrete that cause delamination of coatings.
- k) Penetrates and provides concrete with complete resistance to entry or release of liquids and gases (of heavier atomic weight than oxygen).
- l) Significantly reduces dusting—makes cleaning and sweeping easier.
- m) Prevents entry of carbon dioxide and consequent breakdown of concrete matrix by chemical reaction of carbonation—preventing corrosion potential of reinforcing.
- n) Hardens concrete with substantial increases in abrasion resistance throughout the depth of penetration.
- o) Enhances traction or “non-slip” qualities of concrete surface.
- p) Environmentally friendly, non-toxic and safe to use—no end products damaging to environment.
- q) Concrete hydrates faster and is optimized for maximum development of compressive and flexural strengths in-situ.

4. PACKAGING

- 4.1. Aquaron CPT-2000 is available in 5 gallon and 55 gallon drum sizes, and is also available in bulk container quantities.

- 4.2. The rate of application is 150-200 sq. ft. per U.S. gallon on fresh concrete. The surface must be properly finished to industry standards. Aquaron CPT-2000 can be applied when the surface is damp and free of standing water and bleed water.
- 4.3. For newly cured existing concrete surfaces, all oil, grease, dust, dirt and other deleterious materials must be removed prior to application. Any caustics or degreasers used to remove the deleterious materials must be neutralized and properly cleaned from the surface of the concrete prior to the application of Aquaron CPT-2000. After cleaning the concrete surface, allow time for it to dry to an saturated surface dry condition.
- 4.4. Aquaron CPT-2000 can be applied to damp surfaces, i.e. saturated surface dry (SSD) conditions, must be free of ponded water.
- 4.5. Sealing effectiveness of the Aquaron CPT-2000 is not impaired by abrasion damage to the concrete surface.

5. APPLICATION PROCEDURES

5.1. For Fresh Concrete:

- a) Do not proceed with application of Aquaron CPT-2000 when ambient temperature and/or substrate temperatures are less than 37°F or forecasted to drop below 37°F during the next 6 hours.
- b) Should temperatures be extremely hot, cool surface with sufficient water to decrease temperature, allow surface to achieve a saturated dry surface condition prior to application.
- c) Caution should be taken in windy conditions as Aquaron CPT-2000 if allowed to dry on glass can etch. (Flush immediately with water should contact with glass occur.)
- d) As a water borne water Aquaron CPT-2000 must have access to the surface, any previous coatings or heavy build-ups of oil or grease must be removed.
- e) When utilizing Aquaron CPT-2000 as a cure and seal agent for fresh concrete, a minimum 28-day period must elapse prior to installation/ application of impermeable coats. A test patch for the subject area is recommended.
- f) Coverage rate of 150 sq. ft. per gallon.

5.2. For Established Cured Concrete:

- a) Coverage rate is from 150-200 sq. ft. per gallon.
- b) Application to established cured concrete must be made using a medium to high pressure airless paint sprayer with a .019, 25 degree fan spray tip.
- c) Pre-wet with tap water area of application to dampen, all pooled and puddled areas must be removed. Continue to keep area damp until Aquaron CPT-2000 has been applied.
- d) Always start application at lowest possible area first and proceed to the higher elevations. On vertical applications (walls) start at bottom and proceed up the surface.
- e) When application is to weeping hydrostatic walls or floors, two coats are recommended with second coat being applied immediately following first coat.
- f) On existing concrete, allow a minimum of 72 hours after application of the Aquaron CPT-2000 before installing impermeable coatings and adhesives.

6. DRYING TIME

- 6.1. Initial drying time is approximately 1-2 hours at 70°F, however drying time are directly related to temperature and relative humidity. It is recommended that a minimum of 72 hours be allowed for maximum full drying and penetration of the product prior to any topical application.

7. LIMITATIONS

- 7.1. It is recommended that the product be not applied to a surface with a temperature below 37°F.
- 7.2. Shelf life of the product is indefinite, provided containers are kept tightly sealed when not in use.
- 7.3. If the product comes into contact with glass, immediately flush affected glass with water, or etching of glass will result. It is recommended all glass areas be tarped and taped off to prevent damage.

8. TECHNICAL DATA/ ASTM STANDARDS

- 8.1. **ASTM C672 "Scaling Resistance After Salt Exposure":** Result—no scaling.
- 8.2. **AASHTO T259-80 "Chloride Iron Penetration to Steel":** Result—93% more effective than control.
- 8.3. **ASTM C666 "Resistance to Rapid Freeze-Thawing":** Result—99% more effective than control.
- 8.4. **ASTM C309 "Class 'A' Curing Compound":** Result—0.028 grams per square centimeter. Test requirement not to exceed 0.055 grams per square centimeter. Test period 72 hours. Outperforms other known curing agents.
- 8.5. **Comparative Abrasion Resistance Taber Abrasion Test Method 1,000 Gram Load for 1,000 Revolutions:** 15% increase in abrasion resistance over untreated samples for 35 MPa concrete.
- 8.6. **ASTM C192-88 "Making and Curing Concrete Test Specimens in the Laboratory (Modified)":** Result—Aquron 2000 Concrete Preservation Treatment product increased the strength gained by 8-10% over a 28-day period.
- 8.7. **Concrete Permeability Testing, Pressure Cell Method (Modified), Corps of Engineers (CRD, C-48-73):** Result—Excellent, lower permeability than control specimen (35 MPa concrete). Water did not pass through test specimen over duration of test period.

Prepared by:

GORDON SPRATT & ASSOCIATES LTD.

Per:



Don E. Lee
Technician
DEL/as

GORDON SPRATT & ASSOCIATES LTD.

CONSULTING PROFESSIONAL ENGINEERS
2348 Yukon Street, Vancouver, British Columbia
V5Y 3T6 Phone (604) 872-1211
Fax (604) 872-1274

Our File No. M-499-97

November 28, 1997

MARKHAM DISTRIBUTING
309 East Avenue Road
P.O. Box 2182
Hastings, New Zealand
(Fax: 011-646-878-5143)

Attention: Mr. Graham Smith

Dear Sir:

Re: AQURON PRODUCTS
- Aquaron 2000 Concrete Preservation Treatment
- Professional Opinion Letter

Gordon Spratt & Associates Ltd. were requested to review the following tests/ testimonials conducted with the Aquaron 2000 Concrete Preservation Treatment product:

- a) Test Results American Masonry Protection Inc., U.S.A.
- b) Testimonial from Orlando Utilities Commission, Orlando, Florida.
- c) Marketpro Products Inc., West Vancouver, British Columbia, Canada, Hydroelectrical Utility.

Our professional opinion on the Aquaron 2000 Concrete Preservation Treatment is based on the test results we have obtained from our extensive testing performed in our laboratory and review of the above test results/ testimonials.

We personally met with Mr. Keith Margetson of Marketpro Products and reviewed the photographs of the wall of the power house where serious water leakage problems were occurring. After this area was treated with the Aquaron 2000 Concrete Preservation Treatment system, the first time, the photographs showed minor weeping of the original area, and after the second application the photographs showed that the area was completely dry and was repainted.

Gordon Spratt & Associates Ltd. is satisfied that the product did decrease the permeability of the paste (cement and water) resulting in stopping the water leakage. The permeability of concrete also refers to its ability to resist liquids, gases and chloride ion penetration. Our testing program indicates that the product is capable to decrease the permeability of concrete.

Our File No. M-499-97

After reviewing both the Orlando Utilities Commission letter and American Masonry Protection test results, in conjunction with our extensive test program and use of the product, it is our opinion that the Aquaron 2000 Concrete Preservation Treatment, if applied correctly, will enhance the performance of the concrete as stated in these particular situations.

If there are any questions with regards to the above, please do not hesitate to contact me.

Yours truly,

GORDON SPRATT & ASSOCIATES LTD.

Per:



Don E. Lee
Technician
DEL/as
Encls.

AMERICAN MASONRY PROTECTION, INC.

AQURON™ CPT 2000

CONCRETE PRESERVATIVE/SEALANT TREATMENT

TESTING REPORT ASCERTAINING AQURON™ CPT 2000 MEETS OR EXCEEDS THE MINIMUM REQUIREMENTS OF THE FOLLOWING DIN STANDARDS

DIN-1048 WATER PENETRATION

This report originated at a laboratory for chemical and physical material testing that is a state-authorized testing institute for construction materials.

This report is concerning the basic characteristics and effectiveness of the product, **AQURON™ CPT-2000**.

Applicant: American Masonry Protection, Inc.

P.O. Box 758

Rockwall, Texas 75087 (USA)

Subject:

The applicant manufactures a concrete restoration and protection product on a silicate base under the name of **AQURON™ CPT-2000**. This product is used for insulating and restoring concrete and mortar and with many other inorganic construction materials and structures.

Product samples:

To conduct the tests the applicant has provided us with three product samples. These samples have been tested by us for basic product characteristics on mortar and concrete.

Testing program:

The main results will be given by the different tests concerning water penetration according to DIN 1048, which is to be measured at a water pressure of one (1), three (3) and seven (7) bar.

The concrete samples to be used in the test are to be prepared according to DIN 1045. The aggregate used was a mixture of sand which had previously been tested according to DIN 4226. After that the aggregate was washed, dried then separated into the different sizes of the sand/gravel mix. Then a mixture of the different sizes according to Fuller was calculated and prepared in order to make sure an even concrete structure for all tests which is necessary to compare the results of the different test series.

The following concrete has been prepared for the samples:

VB 1 - (Versushs-Beton = concrete for tests)

Sieve line according to Fuller, prepared with 300 kg portland cement DIN 1164-PZ 35 F per cubic meter of hardened concrete.

(Note: PZ 35 F = portland cement, rapid hardening, causing after 28 days of curing a compressive strength of $35 \text{ N/mm}^2 = 5,000 \text{ psi}$)

VB 2 - Same as VB 1, but with blast furnace cement HOZ 35 L = moderate hardening

VB 3 - Same as VB 1, but with portland cement PZ 45 F - rapid hardening, minimum compressive strength after 28 days $45 \text{ N/mm}^2 = 6,400 \text{ psi}$, used for concrete of strength class B 45 - minimum compressive strength $45 \text{ N/mm}^2 = 6,400 \text{ psi}$, series compressive strength - minimum of three following samples - 7,100 psi.

VB 4 - Same as VB 3, but with blast furnace cement HOZ 45 L = moderate hardening.

The four different samples have been manufactured according to DIN 1045 and controlled according to DIN 1048. These carefully prepared and absolutely compatible mixtures have been used for different tests. The necessary water-plates as well as the samples to test the concrete quality have been manufactured according to DIN 1048. (The dimensions of the water-plates were 200 mm x 200 mm x 120 mm of the samples 300 mm sides).

The surfaces of the water-plates that were to be exposed to water pressure were prepared according to DIN 1048. The test specimens were then placed for 28 days in water (18 - 20 degrees C), then taken out of the thermal bath and the prepared surfaces were treated with **AQURON™ CPT-2000**, as per manufacturers specification. After the first application was dry, this procedure was then repeated.

Then the test specimens were tested for water impermeability according to DIN 1048 (see separate explanation). The specimens were subject to water pressure of one bar for 48 hours, three bar for another 24 hours and finally seven bar for another 24 hours. (14.5 psi/43.5 psi/101.5 psi = 33.5 ft water/100.5/234.5)

After that the water-plates were taken out of the testing-apparatus and split. At the split surfaces the depth of water penetration was immediately measured and marked with a special marker.

List of Average Depth of Water penetration

CONCRETE TYPE NO.	DEPTH OF WATER WITH AQURON™	PENETRATION IN mm WITHOUT AQURON™
1	5.7	26.5
2	8.8	23.0
3	4.2	20.0
4	6.5	21.5
Average	6.3	22.75

This list of average values of all testing series makes it obvious with all four types of concrete tested, that **AQURON™ CPT-2000** reduces the depth of water penetration considerably and therefore insulates as manufacturer claims.

The result of waterproofing depends to a large degree on correct application with a certain amount of care. Therefore the application should only be done by trained contractors, that are willing to warranty their job.

REMARKS:

- The test according to DIN 1048 is used to determine special qualities of concrete according to 1045.
 - waterproof concrete: maximum depth of water penetration no more than 50 mm = 2 inches.
 - concrete with high frost resistance: has to be waterproof, water/cement ratio under 0.6 or air entrained.
 - concrete with resistance to de-icing salts: has to be waterproof, water/cement ratio under 0.5, air entrained and minimum compressive strength required.
 - concrete with high resistance to chemical attack: has to be waterproof, if subject to strong chemical attack maximum depth of water penetration 30mm
- The lab has also established the following values on compressive strength for treated specimens (5000 psi design):

VB 1: 47.2 N/mm² = 6,720 psi
 VB 2: 44.7 N/mm² = 6,370 psi

VB 3: 50.9 N/mm² = 7,250 psi
 VB 4: 54.5 N/mm² = 7,750 psi



ORLANDO UTILITIES COMMISSION

500 SOUTH ORANGE AVENUE • P. O. BOX 3193 • ORLANDO, FLORIDA 32802 • 407/423-9100

Mr. Graham Smith
Markham Distributors
New Zealand

September 3, 1997

The Orlando Utilities Commission (provides electricity and water to City of Orlando, Florida) in 1993 constructed a new office building of 65,000 sq. ft. that had Collins & Aikman carpet squares installed in it. The vinyl backing on this carpet had a mixture problem that led to a failure of the plastic: the plasticizer began to migrate out of the mixture, a combination C-9 and C-10 alcohol provided by the BASF Chemical Company. This material moved both up into the air and down through the carpet adhesive into the concrete sub-flooring. The air of the offices was reading in the 3,000 to 5,000 microns of material per cubic meter, and the concrete had alcohol down to 1-1/2 inches in depth and was emitting back into the air at rates of 6,000 to 7,000 microns. Industrial hygienists stated that any level *above 500* was *unacceptable*, and the alcohol was definitely the contributing factor to occupant complaints of headaches and dizziness.

To keep a long story short, the building had to be vacated, furniture removed, the interior walls and ceiling tiles discarded due to alcohol absorption. In order to remove the alcohol, and to ensure that there would be no problems in the future, a nationwide search was undertaken to find products that would seal the alcohol in the concrete. Many materials were tested on the surface: epoxy, sealers, and concrete treatments. The goal of the testing was to see if it was possible to reduce the emissions from the concrete to acceptable levels prior to replacement of finish flooring material. All of the 20 plus tests failed, with most of the tested materials actually increasing the level of emissions, except one product, **AQURON 2000**. This product was presented as being able to purge salts from concrete while sealing it. Tests showed that it did purge the salts, which appeared as crystal on the surface of the concrete, and along with the purging action *lifted the alcohol to the surface as well*. Core tests showed that the **AQURON 2000** penetrated to 3" in depth with a double coating (applied 20 minutes apart) and further, it sealed the concrete against moisture penetration. Emissions coming from the treated concrete *was reduced to levels of between 50 and 100 microns!*

It was decided that the first floor concrete would be removed for psychological reasons in order to tell our employees that we had done everything possible for their health, but the second floor system was part of the structural diaphragm of the building designed to resist hurricane winds and could not be removed. It was treated with **AQURON 2000** only. *After treatment, the levels of the entire second floor were below the 100 microns. The first floor, where the new concrete was poured, was treated with AQURON 2000 for moisture sealing prior to installation of 12" x 10" vinyl floor tiles (20,000 sq. ft.) with the remainder being covered with Millken Carpet, 36" x 36" carpet sections. The building has been reoccupied without complaint for over a year and a half and passed a thorough Indoor Air Quality test only last month.*

I was the Project Engineer on the construction of this building and consequently the person responsible for the fix. At first I thought the Aquaron product to be a salesman's "snake oil!" I was wrong. **AQURON 2000** proved by test and by experience to be everything it was touted to be, and more.

Sincerely,

Artie J. Spitzer
Sr. Engineer, Facilities Division





ORLANDO UTILITIES COMMISSION

500 SOUTH ORANGE AVENUE ▪ P. O. BOX 3193 ▪ ORLANDO, FLORIDA 32802 ▪ 407/423-9100

Kenneth D. Solomon, President
Aquron Corporation
P.O. Box 758
Rockwall, Texas 75087

Dear Ken,

Let me put in writing the results we have experienced with the **Aquron CPT-2000** product.

The Commission has a new Engineering Office Building at 6003 Pershing Avenue, that has had a carpet failure in which the plasticizer of the carpet backing decomposed into its original chemical make up and released an assortment of chemicals in their gas state, some went into the air and some were absorbed into the concrete under the carpet. These chemicals were mainly harmless but there were a few alcohols that were very troublesome, especially the nonanol, heptonol, pentanol and hexanol. These compounds caused the employees to develop problems typical of sick building syndrome, headaches and upper respiratory problems that were serious enough to force the evacuation of the building and the removal of the carpet.

We conducted many experiments with floor sealers and found that none worked as well as the Aquron CPT-2000. In the tests of epoxy coatings we found that the off-gassing of these products equaled or surpassed the amounts of remaining carpet gasses, but with the Aquron products' off-gassing was nil while the carpet chemicals were decreased to below the levels of smell and in most cases undetectable in testing chambers.

Our chief environmental consultant on this project was GZA GeoInvironmental of Grand Rapids Michigan who used Air Quality Sciences Laboratories of Atlanta to run the chamber tests. Basically the laboratory would prepare a set of absorbent tubes of various charcoal medium through which we would draw air from a aluminum tent placed over the bare concrete or sample floor sealers that had been applied to the floors in small quantities. The exposed tubes would be hooked up to a gas chromatograph-mass spectrometer at the lab for analysis. Without a doubt the Aquron CPT-2000 was superior to the other products.

Even though the Aquron product was not designed to purge these particular chemicals from concrete, the fact remains that it does, as our tests have shown. We are a quasi-government organization and as such, test results are open to the public, please feel free to have any of your clients phone me at (407) 423-1900 if they have questions.

Sincerely,

Anne J. Spitzer
Sr. Engineer, Facilities Division

a:\cmsjerry AJS 3/25/96 aquron 4/1/96



MARKETPRO

Mr. Ken Solomon
American Masonry Protection Inc.
PO Box 758
Rockwall TX 75087

Dear Ken:

The following is a brief outline of a test we performed using Aquaron.

The test was performed at a small dam operated by BC Hydro, the electric utility for our province. They were having serious leaking problems at the powerhouse just below the main dam area. The whole dam site is 75 years old. Half way up a 40 - 50 foot high wall, where the second pour met the first, there was significant leaking along a 40 foot horizontal area. The wall was significantly below ground level and, with the high water table, it was leaking ground water continuously. The dam is fairly high up in the mountains where it rains a lot.

We sprayed the leaking area with Aquaron using a high pressure applicator. The concrete had cracks and holes. There was water leaking out all over the area and we couldn't even tell where we had sprayed.

After about 3 or 4 days we talked to the maintenance people at the dam who informed us that the leaking had slowed but the area was still pretty bad. We thought we must have sprayed incorrectly so made arrangements to go back up and respray.

By the time we went back about two weeks later, there was a significant reduction in the leaking. Some areas had stopped altogether and other areas were only wet where there had once been water pouring through. The maintenance people informed us that each day there was less water leaking through. They were astounded.

We sprayed for a second time and inspected the site again after a further two week period. This time the total area had all but stopped. We went back to do a final inspection about two months later. The whole area had been repainted and there was no indication at all that there had ever been a problem with leaks!

We are about to do some more problem area for them and will keep you informed as to the results.

Sincerely



Keith Margetson,
President

ASTM C-114

RE: Aquoron CPT-2000 as a cure method-ASTM C-114 Test.

Excerpted from a document originating at SUMMIT TESTING & INSPECTION CO. OF AKRON UNIVERSITY, AKRON, OHIO

“This letter represents a summary of testing services provided relative to determining water soluble chloride ions in various concrete specimens.

The testing was performed on a concrete parking garage deck being built for a University Performing Arts Hall. Our company was providing construction inspection and testing to this project.

Initially, samples for chloride ion testing were hammer drilled from space test cylinders cast from the delivered concrete.

Subsequent test specimens were drilled after 12 months from the top down to 75-100mm in the areas of the parking deck where the concrete used to make the cylinders was placed. The deck had been treated with Aquoron CPT-2000 for rebar protection by the contractor.

Between placing and finishing of the concrete, and the initial samples and the samples extracted 12 months later. The carpark surface was repeatedly exposed to road salts applied to reduce ice (for traffic safety).

Test samples were taken to a depth of 75 to 100mm deep into the concrete, proving the penetration of Aquoron CPT-2000 and an effective reduction in chlorides as well as forming a barrier to chloride ingress.

Actual testing was done by another laboratory through agreement with ours.”

RESULTS:

WATER SOLUBLE CHLORIDE ION*

Pour Date	% of Total Sample		Calculated % of Cementious Fraction	
	1990	1991	1990	1991
07/31/1990	0.035	0.026	0.21	0.19
	0.026	0.002	0.19	0.012
8/01/1990	0.029	0.003	0.21	0.018
	0.024	0.007	0.18	0.042
8/23/1990	0.032	0.002	0.24	0.012
	0.041	0.013	0.30	0.076
8/24/1990	0.032	0.003	0.24	0.018
	0.043	0.002	0.32	0.012

*by potentiometric Titration with Silver Nitrate (modified ASTM C-114)

REPORT & CONCLUSION

Based on these results, the AQURON CPT-2000 appears to not only prevent an increase of chloride ion content in concrete exposed to de-icers and vehicular traffic, but also has reduced the chlorides present in the concrete when it was produced.

SUMMIT TESTING & INSPECTION COMPANY
JOHN MALIVUK, PE



Summit Testing & Inspection Company

P.O. Box 2231, Akron, Ohio 44309
Deliveries: 910 White Pond Dr., Akron, Ohio 44320
(216) 869-6605

November 23, 1992

American Masonry Protection
Box 758
Rockwall, TX 75087

Attention: Ken Solomon

This letter presents a summary of testing services provided relative to determining water soluble chloride ions in various concrete specimens.

The testing was performed on concrete being supplied to the University of Akron (Ohio) for repairs to the E. J. Thomas Performing Arts Hall Parking Deck. Our company was providing construction inspection and testing to this project.

Initially, samples for Chloride Ion testing were hammer drilled from spare test cylinders cast from the delivered concrete. Subsequent test specimens were drilled after 12 months ± from the top 3-4 inches in the areas of the parking deck floor where the concrete used to make the cylinders was placed. The deck had been reportedly treated with AQUARON 2000 cure and seal.

The actual testing was done by Construction Technology Laboratories, Inc. The results are:

WATER SOLUBLE CHLORIDE ION *

<u>Pour Date</u>	<u>% of total sample</u>		<u>Calculated % of Cementitious Fraction</u>	
	<u>1990</u>	<u>1991</u>	<u>1990</u>	<u>1991</u>
07/31/90	0.026	0.035	0.19	0.21
	0.026	0.002	0.19	0.012
08/01/90	0.029	0.003	0.21	0.018
	0.024	0.007	0.18	0.042
08/23/90	0.032	0.002	0.24	0.012
	0.041	0.013	0.30	0.078
08/24/90	0.032	0.003	0.24	0.018
	0.043	0.002	0.32	0.012

American Masonry Protection
November 23, 1992
Page 2

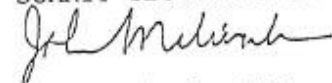
*By Potentiometric Titration with Silver Nitrate (Modified
ASTM C114

Based on these results, the AQUORON 2000 appears to not only prevent an increase of chloride ion content in concrete exposed to de-icers and vehicular traffic, but also reduce the amount already present.

Should there be any questions regarding these tests, please do not hesitate calling our office.

Respectfully submitted,

SUMMIT TESTING & INSPECTION COMPANY



John Malivuk, P.E.

JM:p

PC: file

Aquron Corporation International
2009 Industrial Blvd # A
Rockwall, TX 75087

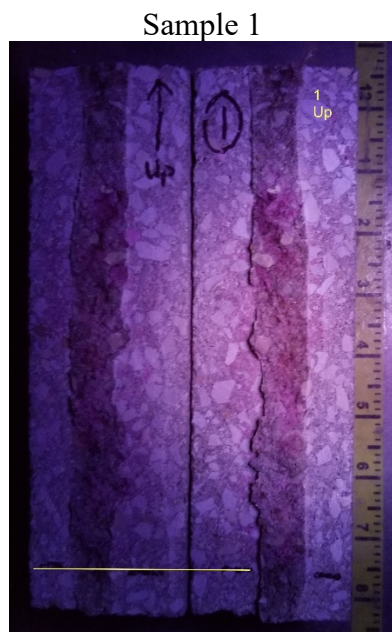
May 24, 2020

Re: **Aquron 2000 Penetration Testing**
Omaha, NE
Project No. 72-61

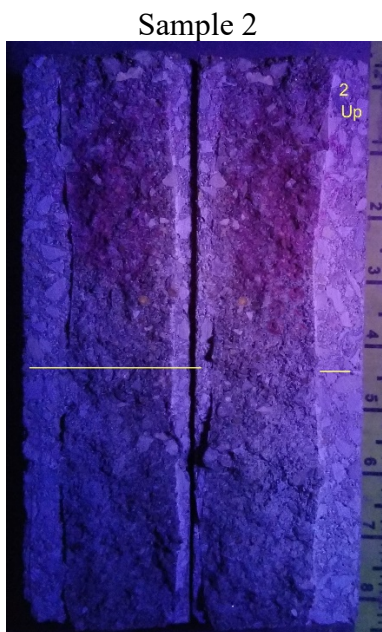
On May 23-24, 2020, ISG-TEAM performed Fluorescent Penetration testing on three solid concrete masonry unit block samples measuring 3.75" x 2.25" x 7.63". A summary of the test method:

The samples were prepared so applied solution would contact the top surface only by shrink wrapping the sides with non-absorbent sheet and sealed with metal tie wire.

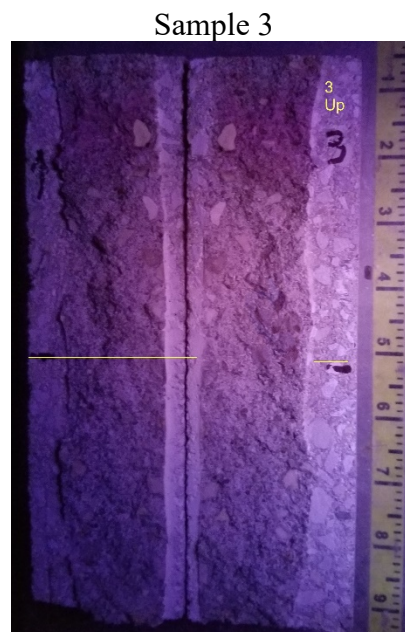
A solution was prepared by adding reagent grade Potassium Permanganate (dye) to Aquron 2000 Cure & Seal. Solution was applied in four to five sprayed perpendicular passes until a glossy sheen was observed. 40 hours after the solution was applied, the samples were split lengthwise and the penetration was observed under a black light. Observed penetration as follows:



Penetration: 7.1"



Penetration: 4.9"



Penetration: 5.0"

Average Penetration: 5.7"

Please contact us if you have any questions.

Jeff Roberts, President